

Your slide-by-slide pitch guide

What to say on every slide, the numbers to know cold, the questions a jury or investor will ask, and how to answer them honestly.

15 slides	~10 min full pitch, plus Q&A	₹16 lakh the ask: 9 months to PoC + raise	1 goal a working proof of concept
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How to use this guide

- Read each slide page once, then say the script **out loud** three times. Don't memorise it word for word; memorise the **one thing they must remember** and the numbers.
- Practise the Q&A with a friend asking the questions in random order. Answer in **two or three sentences**, then stop.
- Use the 10-minute running order (next page) with a timer. If you run long, use the 5-minute cuts.
- Before the pitch, settle the open decisions on the "Fix or decide before you present" page. A juror who spots an inconsistency stops listening to the story.

Your three golden rules

Rule	What it means on stage
1. Say "projected" for every future number	Unit costs, prices, margins, market plans and the seed round are plans. Only the prototype BOM is costed today.
2. Be honest about stage	Baker: raw prototype running. Mixer: architecture defined, not built. Proto: inside Baker Studio. Honesty is your strongest slide (09).
3. Every answer ends on the loop	Whatever the question, come back to: design to tested board the same day, instead of weeks.

10-minute running order

#	Slide	Time	Cumulative	5-minute version
1	Cover · Focus on innovation	0:20	0:20	Keep (0:15)
2	01 Executive summary	0:45	1:05	Keep (0:30)
3	02 Problem	0:50	1:55	Keep (0:35)
4	03 Solution	0:50	2:45	Keep (0:35)
5	04 Proto	1:00	3:45	Keep (0:40)
6	05 A new category	0:35	4:20	One line only
7	06 Competition	0:40	5:00	Keep (0:25)
8	07 Business model (projected)	0:50	5:50	Keep (0:30)
9	08 Market plan	0:45	6:35	One line: “start in maker studios”
10	09 Where we are	0:40	7:15	Keep (0:25)
11	10 What the money proves	0:45	8:00	Keep (0:30)
12	11 The ask	0:35	8:35	Keep (0:25)
13	12 Cost breakup	0:30	9:05	Skip: “details are in the deck”
14	13 What’s next	0:45	9:50	One line: seed round + China
15	Thank you	0:15	10:05	Keep (0:10)

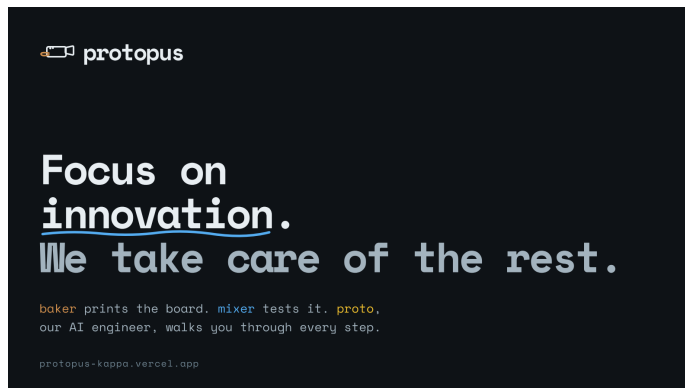
The 5-minute spine

If the jury gives you **5 minutes**, the story is: problem (the loop takes weeks) → solution (Baker, Mixer, Proto) → why ours (Proto + one desk) → competition (nobody closes the loop) → projected model → honest progress → milestones → ₹16 lakh. Everything else is Q&A material.

Before you walk in

- Demo video of the raw Baker prototype downloaded **offline**, on the laptop and on your phone.
- Website open in a tab: protopus-kappa.vercel.app (and a screenshot in case the Wi-Fi fails).
- The costing workbook open on the PoC Budget sheet, for “show me how you got ₹16 lakh”.
- One printed copy of the deck and of the numbers cheat sheet (last pages of this guide).
- Water. Speak slower than feels natural: the slides are dense, give people time to read.

Slide 1 · Cover



On screen

Focus on innovation. We take care of the rest.

Time

20 seconds

The one thing they must remember

Protopus is three things: Baker prints the board, Mixer tests it, Proto walks you through every step.

What to say

“Hi, I’m Soham, a third-year ECE student at the Faculty of Technology, University of Delhi. This is Protopus.”

“Our line is simple: focus on innovation, we take care of the rest. Baker prints the board. Mixer tests it. And Proto, our AI engineer, walks you through every step.”

Delivery tip

Pause for one beat after “We take care of the rest.” Point at the three coloured words as you say them.

Questions you may get

Q What is Protopus in one sentence?

A desktop PCB factory, a full test bench and an AI engineer, so one person can go from a design to a tested board the same day instead of weeks.

Q Is Protopus a company yet?

Answer with your actual status. If not incorporated yet: “We will incorporate in this phase; the IP and legal line in the budget covers it.”

Slide 2 · 01 · Executive summary

01 · EXECUTIVE SUMMARY			
A small electronics factory and a full test bench, on one desk			
MORNING You draw it Proto checks the design and points at what will bite later.	MIDDAY Baker builds it Board printed, parts placed and soldered.	AFTERNOON Mixer tests it One cable, eight instruments, a reading you can trust.	EVENING You try again The next attempt is tonight, not two weeks from now.
WHAT MAKES IT OURS Making and testing on one desk, with Proto guiding every step instead of a hundred buttons.			

On screen

A small electronics factory and a full test bench, on one desk

Time

45 seconds

The one thing they must remember

The whole loop, design to tested board, happens in one day on one desk.

What to say

“Let me walk you through a day with Protopus.”

“Morning: you draw your circuit, and Proto checks the design and points at what will bite you later.”

“Midday: Baker builds it. The board is printed, the parts are placed and soldered.”

“Afternoon: Mixer tests it. One cable, eight instruments, a reading you can trust.”

“Evening: you try again. The next attempt is tonight, not two weeks from now.”

“What makes it ours is putting making and testing on one desk, with Proto guiding every step instead of a hundred buttons.”

Numbers to know cold

- 1 desk · 1 cable · 8 instruments
- Next attempt: tonight, not in two weeks

Questions you may get

Q “Same day”: how long does one board actually take?

Be honest: “Same-day is the design goal. Our month-2 milestone is Baker running a full job hands-off on video, and that gives us the real cycle time.” Do not quote hours you have not measured.

Q Is placement automatic?

No, and that is deliberate. The operator drops each part into a 3D-printed jig with a foot-pedal vacuum pen; the jig does the aiming. Soldering is automatic reflow on the hotplate. We removed cameras and feeders to keep the cost down.

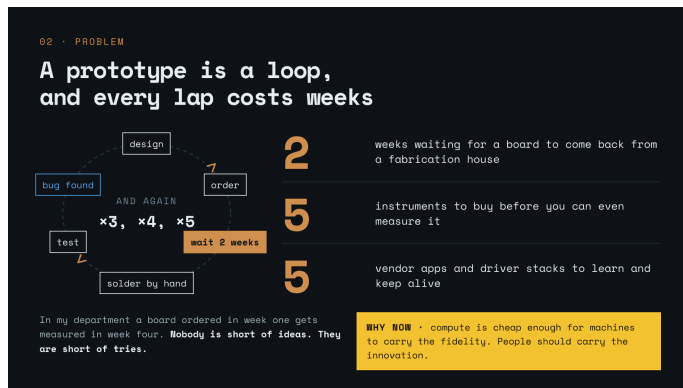
Q What does Proto actually check?

It reads the design, flags problems like a trace that is too thin, fixes it with your approval, and plans the machine job. Slide 04 shows a real example.

Avoid saying

- “Fully automatic” or “pick-and-place robot”: placement is manual, with a jig.

Slide 3 · 02 · Problem



On screen

A prototype is a loop, and every lap costs weeks

Time

50 seconds

The one thing they must remember

Nobody is short of ideas. They are short of tries.

What to say

“Every hardware prototype is a loop: design, order, wait two weeks, solder by hand, test, find a bug, and go again. Three, four, five times.”

“In my department, a board ordered in week one gets measured in week four.”

“And before you can even measure it, you need about five instruments, and five vendor apps and driver stacks to learn and keep alive.”

“Nobody is short of ideas. They are short of tries.”

“Why now? Compute is finally cheap enough for machines to carry the fidelity, so people can carry the innovation.”

Delivery tip

The department story is your strongest moment. Say it slowly, like a memory, not a statistic.

Numbers to know cold

- 2 weeks waiting for a board from a fab house
- 5 instruments to buy before you can measure
- 5 vendor apps and driver stacks
- Loops: x3, x4, x5 per project

Questions you may get

Q JLCPCB is cheap and fast now. Why is this a problem?

Cheap per board, yes. But from India it is 1–3 weeks plus customs, and every bug restarts that clock. We are not competing on board price; we are competing on loop time.

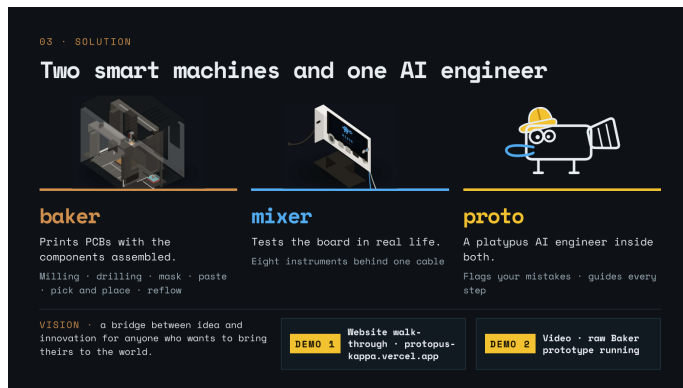
Q Who exactly feels this pain?

Students and college labs, makers, hardware startups and school tinkering labs. Slide 08 sizes them.

Q Is your department story typical?

Say what you have seen: “It is our experience in the lab, on every project.” If you have spoken to other colleges or startups, say how many.

Slide 4 · 03 · Solution



On screen

Two smart machines and one AI engineer

Time

50 seconds

The one thing they must remember

Baker makes the board, Mixer tests it, Proto guides both.

What to say

“Our solution is two machines and one AI engineer.”

“Baker prints PCBs with the components assembled: milling, drilling, solder mask, paste, pick and place, and reflow.”

“Mixer tests the board in real life: eight instruments behind one cable.”

“And Proto, a platypus AI engineer inside both, flags your mistakes and guides every step.”

“Our vision is a bridge between idea and innovation for anyone who wants to bring theirs to the world. You can see the website walkthrough and a video of the raw Baker prototype running.”

Delivery tip

Have the demo video cued before the pitch starts. If there is no screen time for it, offer it in Q&A.

Numbers to know cold

- Baker: 15 steps on one shuttle, one coordinate frame
- Mixer (prototype): 2-ch scope 12-bit 65 MS/s · 2-ch waveform generator 14-bit · 16-ch logic analyser · pattern generator · protocol decoder · 2 programmable supplies · 24-bit multimeter
- Board size in the workflow: 100 × 160 mm (see “Fix or decide”)

Questions you may get

Q How does Baker make the board?

A touch probe maps the board height, the spindle mills the traces and drills, a syringe coats solder mask, a 10.1-inch UV screen cures everything except the pads, a laser cuts a stencil in place, paste goes on, you place parts in a printed jig, and a hotplate reflows. One fixture, one coordinate frame the whole way.

Q What class of instrument is Mixer?

Analog Discovery 3 class. The prototype has 12-bit capture so we can tune the analog front end; the production version drops to 8-bit to reach the ₹15,000 hardware target.

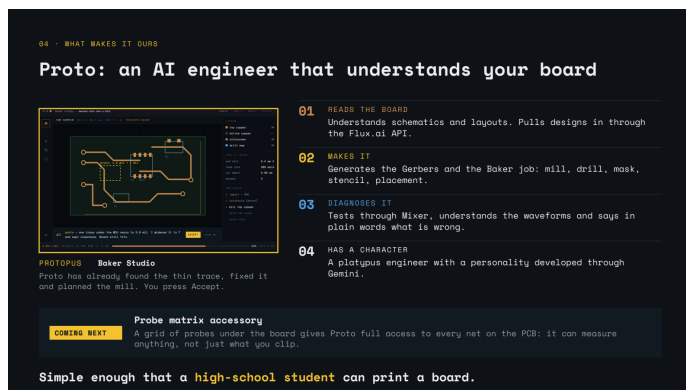
Q Multilayer boards? Bigger boards?

Not in the base machine. Bigger beds and multilayer are on the “industrial variants, later” line.

Q Can I see it working?

Show the Baker video. Be clear it is the raw prototype, and that the full 15-step hands-off run is the month-2 milestone.

Slide 5 · 04 · What makes it ours (Proto)



On screen

Proto: an AI engineer that understands your board

Time

60 seconds

The one thing they must remember

Proto reads the board, makes it, diagnoses it, and has a character people enjoy.

What to say

“What makes Protopus ours is Proto, an AI engineer that understands your board.”

“One: it reads the board. It understands schematics and layouts, and pulls designs in through the Flux.ai API.”

“Two: it makes it. It generates the Gerbers and the Baker job: mill, drill, mask, stencil and placement.”

“Three: it diagnoses it. It tests through Mixer, understands the waveforms, and tells you in plain words what is wrong.”

“Four: it has a character, a platypus engineer whose personality we developed with Gemini.”

“Here in Baker Studio, Proto has already found a thin trace, fixed it and planned the mill. You just press Accept.”

“Next, we are building a probe matrix accessory: a grid of probes under the board that gives Proto access to every net, so it can measure anything, not just what you clip.”

“Simple enough that a high-school student can print a board.”

Numbers to know cold

- Example on screen: a trace necked to 5.8 mil; Proto widened it to 7 mil
- Probe matrix: coming next, not built yet

Questions you may get

Q Are you dependent on Flux.ai and Gemini?

Suggested answer, if true: “We use the best tools available today: Flux.ai’s API to bring designs in, Gemini for Proto’s character. Our value is how Proto connects the design to our machines and to real Mixer measurements. The model providers can be swapped.”

Q Which parts of Proto work today?

Answer exactly: which of the four capabilities run today, which are shown in Baker Studio as a demo, and which are planned. The probe matrix is future work.

Q What if Proto is wrong?

Proto proposes; you accept. The Accept button is there on purpose. Every change is visible, and Mixer measures the real board, so a wrong suggestion shows up in the test.

Q How does Proto “understand waveforms”?

Mixer captures scope and logic data; Proto compares it with what the design expects and explains the difference in plain words.

Avoid saying

- Calling any capability “working” if it is still planned. If unsure, say “in development”.

Slide 6 · 05 · A new category

05 · A NEW CATEGORY

Between the DIY bench and industrial manufacturing

	DIY bench breadboards, hand etching, borrowed scopes	protopus baker + mixer + proto	industrial fab houses, assembly houses, bench
TIME PER LOOP	1-3 days	Same day	1-3 weeks + customs
EXPERTISE	A seasoned engineer	Proto guides you	Expert to design and test
ASSEMBLED	By hand, if you can	Yes, by Baker	Yes, at extra cost
TESTED	Only if you own a bench	Yes, by Mixer	Separate + lakhs of instruments

The DIY bench needs an expert. The factory needs time and money.
Most ideas have neither.

On screen

Between the DIY bench and industrial manufacturing

Time

35 seconds

The one thing they must remember

The DIY bench needs an expert; the factory needs time and money; Protopus needs neither.

What to say

“Today you have two choices.”

“The DIY bench: breadboards, hand etching, borrowed scopes. One to three days a loop, and it needs a seasoned engineer.”

“Or industrial: a fab house and an assembly house. One to three weeks plus customs, you still need an expert, and lakhs of instruments to test.”

“Protopus sits between them: same day, Proto guides you, Baker assembles, Mixer tests.”

“The DIY bench needs an expert. The factory needs time and money. Most ideas have neither.”

Numbers to know cold

- DIY: 1-3 days · industrial: 1-3 weeks + customs · Protopus: same day

Questions you may get

Q Is this a real category or a niche?

A real, under-served middle: 10,000 Atal Tinkering Labs with 50,000 more planned, about 2,500 Fab Labs worldwide, plus every college electronics lab. Slide 08.

Q Who would still choose the industrial route?

Anyone making hundreds of boards. We are for the tries before that: the first 1 to 20 boards of an idea.

Slide 7 · 06 · Competition

06 · COMPETITION

Others solve one step. Nobody closes the whole loop.

PRODUCT	PRICE · ₹	MAKE	PLACE	TEST	GUIDE
Voltera V-One	~3.1 lakh	●	○	○	○
LPKF ProtoMat	8.6-24.5 lakh	●	○	○	○
Bantam Tools mill	~3.1 lakh	●	○	○	○
Digilent Analog Discovery 3	~33,000	○	○	●	○
JLCPCB, PCBWay	per order, weeks	●	●	○	○
protopus baker + mixer	3.34 lakh	●	●	●	●

List prices at ₹88 per US\$, before import duty. Voltera dispenses paste and reflows, but you place every part by hand. voltera.io · Design World · Digilent

On screen

Others solve one step. Nobody closes the whole loop.

Time

40 seconds

The one thing they must remember

Competitors make or test. Only Protopus makes, places, tests and guides.

What to say

“Others solve one step.”

“Voltera, LPKF and Bantam make boards, from about 3 lakh to 24 lakh. Digilent’s Analog Discovery tests them, at about 33,000. JLCPCB and PCBWay make and place, but per order, in weeks.”

“Nobody closes the whole loop. Protopus, Baker plus Mixer, makes, places, tests and guides, for a projected 3.34 lakh.”

Numbers to know cold

- Voltera V-One ~₹3.1L · LPKF ProtoMat ₹8.6-24.5L · Bantam ~₹3.1L · Digilent AD3 ~₹33,000
- Protopus Baker + Mixer: ₹3.34L projected list (₹2,99,000 + ₹34,999)
- Competitor prices: list prices at ₹88/\$, before import duty

Questions you may get

Q Voltera already dispenses paste and reflows. What is different?

With Voltera you still place every part by hand, and you still need separate test instruments. We add solder mask, a stencil cut in place, jig placement, an eight-instrument test bench, and Proto guiding the whole loop.

Q You are more expensive than Voltera.

For about the same money you get the mill, mask, stencil, placement and reflow, plus a test bench that alone costs around ₹33,000, plus Proto. And their price is before import duty.

Q What stops a big player copying you?

Honest answer: nothing permanent yet. Our edge is the closed loop and speed, and the data from real tests makes Proto better over time. We are filing IP in this phase.

Q Why is your ₹3.34L so far above the ₹1.45L production cost?

The ₹3.34L is the projected selling price; assembly, shipping, warranty, software and support sit in between. Slide 07.

Avoid saying

- “We have no competition.” You do; you just close the loop they don’t.

Slide 8 · 07 · Business model (projected)

07 · BUSINESS MODEL · PROJECTED

What we aim for: sell the machines, and one Proto plan runs all of them

PROJECTED PER UNIT	BAKER	MIXER
Hardware (target)	1,50,000	15,000
Assembly and QC	12,000	2,000
Packing and shipping	8,000	1,000
Warranty reserve	7,500	700
Software share	10,000	2,000
Sales and support	12,500	1,750
Projected total cost	2,00,000	22,500
Target selling price	2,99,000	34,999
Projected margin	~33%	~36%

Costed model: Baker ₹1,44,842, Mixer ₹13,740 incl. assembly.

ROADMAP TO THESE NUMBERS

NOW	MONTHS 9-9 - THIS ASK	AFTER POC - SEED	100-UNIT BATCHES
Prototypes costed: Baker ₹5.73L, Mixer ₹76,847	~16L proof of concept: Baker Rev B, 5 Mixers in studios	~1.5 Cr: China pilots, first 100 Mixers + 20 Bakers	The projected unit economics above

PLANNED · ONE PROTO PLAN · EVERY MACHINE
₹9,999 / year
 One subscription per workspace: Baker Studio, Mixer Station, Proto and every update.

Consumables
 Board blanks, paste and end mills, reordered as labs keep building.

Industrial variants, later
 Bigger beds, milliliter and production test come after the base machines.

All unit costs, prices and margins are projections for production volumes, not current figures.

On screen

What we aim for: sell the machines, and one Proto plan runs all of them

Time

50 seconds

The one thing they must remember

Projected: ~33% margin on Baker, ~36% on Mixer, plus a ₹9,999/year Proto plan and consumables.

What to say

“This is what we aim for. These are projected numbers at 100-unit batches, not today’s figures.”

“We sell the machines. A Baker is projected to cost us 2 lakh all-in and sell at 2.99 lakh, about 33% margin. A Mixer costs 22,500 and sells at 34,999, about 36%.”

“Our costed China-parts model already puts hardware plus assembly at 1.45 lakh for Baker and 13,740 for Mixer, inside these targets.”

“On top, one planned Proto plan runs all of them: 9,999 a year per workspace. And labs keep reordering consumables.”

“The roadmap: prototypes are costed now; this 16 lakh buys the proof of concept; a seed round after that funds the first 100 Mixers and 20 Bakers in China; and at 100-unit batches, these economics apply.”

Numbers to know cold

- Baker: target cost ₹2,00,000 → price ₹2,99,000 → ~33%
- Mixer: ₹22,500 → ₹34,999 → ~36%
- Costed production unit: Baker ₹1,44,842 · Mixer ₹13,740 (incl. assembly)
- Headroom vs hardware target: Baker ₹5,158 · Mixer ₹1,260 (thin)
- Proto plan: ₹9,999/year per workspace

Questions you may get

Q 33% is thin for hardware.

It is deliberate, to get machines into labs. Margin grows from the Proto plan and consumables, and from volume pricing on the three parts still priced at retail: spindle, laser and screen.

Q How did you get ₹1,44,842?

Parts from China at 100-unit prices, 16.5% duty, sea freight, and ₹12,000 for assembly and QC in Delhi. 81 of the 96 prototype BOM lines are priced from live listings.

Q Why would a lab pay a subscription?

The plan covers Baker Studio, Mixer Station, Proto and every update: the software that makes the machines easy. One plan per workspace, however many machines.

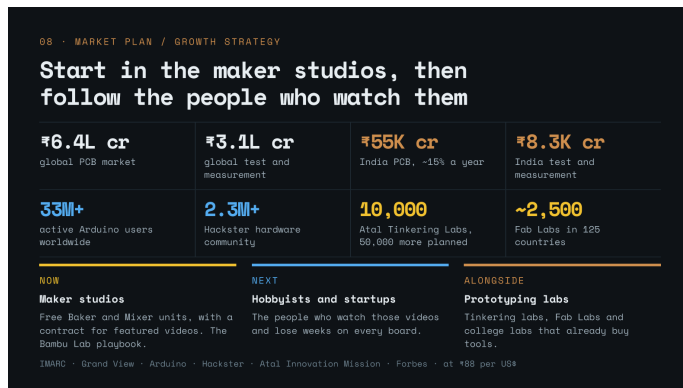
Q What if costs overrun?

Headroom is thin and we know it: Baker is ₹5,158 under target. That is why the next step is volume quotes on the spindle, laser and screen, which today are retail prices.

Avoid saying

- Presenting these as current numbers. Say “projected” and “target” every time.

Slide 9 · 08 · Market plan / growth strategy



On screen

Start in the maker studios, then follow the people who watch them

Time

45 seconds

The one thing they must remember

Start with maker studios (the Bambu Lab playbook), then their audience, alongside labs that already buy tools.

What to say

“The market is large: a 6.4 lakh crore global PCB market and 3.1 lakh crore in test and measurement. In India, 55,000 crore of PCBs growing about 15% a year, and 8,300 crore of test and measurement.”

“The people are already there: 33 million active Arduino users, 2.3 million on Hackster, 10,000 Atal Tinkering Labs with 50,000 more planned, and about 2,500 Fab Labs in 125 countries.”

“Our plan: start in the maker studios, with units in exchange for featured videos. That is the Bambu Lab playbook.”

“Next, the hobbyists and startups who watch those videos and lose weeks on every board. And alongside, the tinkering labs, Fab Labs and college labs that already buy tools.”

Numbers to know cold

- Global: PCB ₹6.4L cr · T&M ₹3.1L cr
- India: PCB ₹55K cr (~15%/yr) · T&M ₹8.3K cr
- 33M+ Arduino users · 2.3M+ Hackster · 10,000 ATIs (+50,000 planned) · ~2,500 Fab Labs in 125 countries
- Sources on slide: IMARC, Grand View, Arduino, Hackster, Atal Innovation Mission, Forbes; ₹88/\$

Questions you may get

Q That is TAM. What is your realistic market?

Start with labs: if just 1% of today's 10,000 Atal Tinkering Labs bought a Baker + Mixer at the projected ₹3.34 lakh, that is about ₹3.3 crore. It illustrates the entry point; it is not a forecast.

Q Why maker studios first?

They create the videos that sell the product to everyone else, and they give fast real-world feedback. Bambu Lab grew exactly this way.

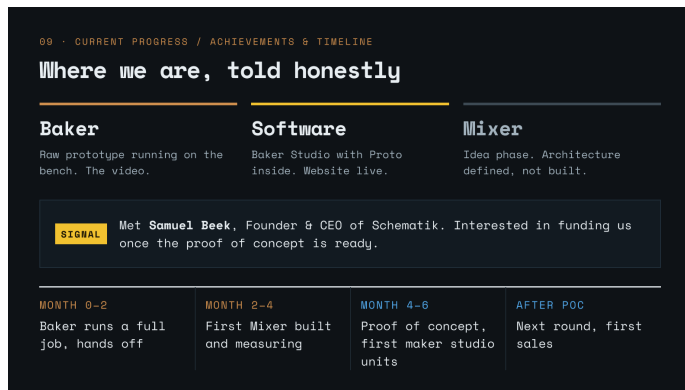
Q Your PoC budget has Mixers for studios but no extra Baker. The slide says free Baker units now.

Prepared answer: “In the proof of concept, the prototype Baker runs in the university lab with students and makers, and 5 production Mixers go to studios. Free Baker units for studios come with the seed round.” (Or change the slide wording first.)

Q Do tinkering labs have budgets?

They receive government grants for equipment through the Atal Innovation Mission. Check the current grant amount before quoting a figure.

Slide 10 · 09 · Where we are



On screen

Where we are, told honestly

Time

40 seconds

The one thing they must remember

Baker prototype runs, the software is live, Mixer is designed, and a founder-CEO is interested once the PoC is ready.

What to say

“Here is where we are, told honestly.”

“Baker: a raw prototype running on the bench; you saw the video.”

“Software: Baker Studio with Proto inside, and our website is live.”

“Mixer: idea phase. The architecture is defined, but it is not built yet.”

“One signal: I met Samuel Beek, Founder and CEO of Schematik. He is interested in funding us once the proof of concept is ready.”

“Our timeline: Baker runs a full job hands-off in months 0 to 2, the first Mixer is built and measuring by month 4, the proof of concept and first maker-studio units by month 6, and then the next round and first sales.”

Numbers to know cold

- PoC by month 6 · next round after

Questions you may get

Q If Baker already runs, why does the ask include ₹3.73 lakh to build Baker?

Prepare this with what your raw prototype really contains. Suggested shape: “The raw prototype proves the core motion and process. The ₹3.73 lakh builds the complete machine: all 15 steps, the safety enclosure, the laser, the UV engine and the high-temperature jig printer.”

Q Is Schematik committed?

No. It is interest, conditional on the proof of concept. That is exactly why the PoC is our goal.

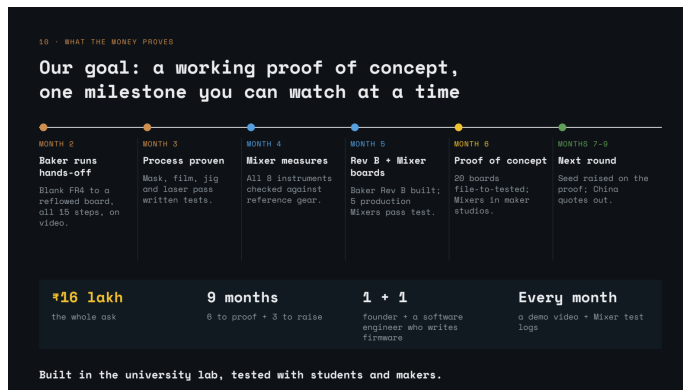
Q Why is Mixer not built yet?

We built Baker first because it is the harder machine. Mixer uses known parts; it is costed at ₹74,847 and is the month-4 milestone.

Avoid saying

- Describing the Schematik interest as an investment or a commitment.

Slide 11 · 10 · What the money proves



On screen

Our goal: a working proof of concept, one milestone you can watch at a time

Time

45 seconds

The one thing they must remember

Six milestones in nine months, each one something you can watch.

What to say

“Our goal is a working proof of concept, and you can watch it happen one milestone at a time.”

“Month 2: Baker runs hands-off, from blank board to reflowed board, all 15 steps, on video.”

“Month 3: the process is proven: mask, film, jig and laser each pass a written test.”

“Month 4: Mixer measures, with all eight instruments checked against reference equipment.”

“Month 5: Baker Rev B and five production Mixer boards.”

“Month 6: the proof of concept: 20 boards taken from file to tested, and Mixers in maker studios.”

“Months 7 to 9: we raise the next round on that proof.”

“Sixteen lakh, nine months, the founder plus one software engineer who writes firmware, and a demo every month.”

Delivery tip

Point along the timeline as you speak; it makes the plan feel concrete.

Numbers to know cold

- 4 process points: 0.5 mm-pitch mask cleaning · film registration $\pm 50 \mu\text{m}$ · jig-to-pin accuracy after resin shrink · laser offset drift with temperature
- 20 boards file-to-tested by month 6 (your target: confirm it)
- 9 months = 6 to PoC + 3 to raise

Questions you may get

Q What if a month-3 test fails?

That is exactly what the Baker Rev B money and the reserve are for. Each point has a fallback, for example skimming the pads with the spindle if the mask does not wipe clean.

Q How will we know you hit a milestone?

Every month ends in a demo video plus the Mixer test logs. You can hold us to each one.

Q Why 20 boards?

Enough boards, made by people other than me, to show the loop works repeatably, not once. (Adjust if you choose a different target.)

Slide 12 · 11 · The ask



On screen

₹16,00,000

Time

35 seconds

The one thing they must remember

₹16 lakh buys nine months: a proof of concept and time to raise.

What to say

“We are asking for sixteen lakh rupees: nine months to a proof of concept, and time to raise the next round.”

“4.48 lakh builds Baker and Mixer, costed line by line. 2.2 lakh covers one iteration: Baker Rev B, five production Mixers and consumables.”

“4.5 lakh is one software engineer who also writes the firmware, for nine months. 1.35 lakh is Proto’s software and compute. 1.5 lakh is testing, tooling and lab safety. And 1.97 lakh covers IP, outreach and a small reserve.”

“I take no stipend. It is enough to survive one failed iteration and raise from strength. This builds the proof; the next round builds the business.”

Numbers to know cold

- Hardware ₹4,48,265 (Baker ₹3,73,418 + Mixer ₹74,847)
- Iteration ₹2,20,000 (Rev B ₹1L · 5 Mixers ₹60k · consumables ₹60k)
- Engineer ₹4,50,000 (₹50,000 × 9)
- Proto software + compute ₹1,35,000 (₹15,000 × 9)
- Lab ₹1,50,000 (testing + tooling ₹1L · lab safety ₹50k)
- IP ₹75k · outreach ₹50k · reserve ₹71,735 (6.6%)

Questions you may get

Q Why not ₹10 lakh?

At ₹10 lakh we build once, with no money to fix what breaks, and the runway ends exactly at the proof of concept, with no time to raise. One failed process point would stop the project.

Q Why no founder salary?

I am a student; every rupee goes into the proof.

Q Why ₹50,000 a month for the engineer?

One person covering both software and firmware; that is roughly the entry-level market in Delhi for that combination.

Q If the grant is capped at ₹10 lakh, what do you cut?

Keep the hardware (₹4.48L), Baker Rev B (₹1L), about 6 months of the engineer, Proto compute, consumables and testing. Defer the production Mixer boards, IP filing, outreach and reserve. The runway drops to about 6 months, so fundraising starts at month 4.

Q Is this grant or equity?

Answer for your situation. The slide says “grant / funding required”; know which you are applying for.

Slide 13 · 12 · Cost breakup

12 · COST BREAKUP

Where the ₹16 lakh goes, line by line

Baker	₹3,73,418	Mixer	₹74,847	Use of funds	₹16,00,000
Frame & enclosure	22,795	Capture + generation	19,138	■ Baker prototype build	3,73,418
Motion	80,382	Front end + instruments	15,762	■ Mixer prototype build	74,847
Zone A: mill, drill, laser	38,270	Host, screen, I/O	29,762	■ Baker Rev B iteration	1,00,000
Zone B: host, UV, wipe	33,058	Enclosure	1,510	■ Mixer production board ×5	50,000
Stencil film	5,805	Parts	62,212	■ PoC consumables	60,000
Placement + reflow	4,891	Import duty + courier	2,872	■ Engineer (software + firmware)	4,50,000
Board handling	300	Contingency (15%)	9,763	■ Proto software + compute	1,25,000
Electronics & control	42,985	Prototype total	₹74,847	■ Testing + tooling	1,00,000
Resin printer (jigs)	77,698			■ Lab set-up + safety	60,000
First-run consumables	0,000	PRODUCTION UNIT (PROJECTED)		■ IP + legal	75,000
Parts	₹3,12,465	Baker	₹1,44,842	■ Demo + studio outreach	50,000
Import duty + courier	12,247	target ₹1,00,000 · under by ₹1,100		■ Reserve (5.0%)	21,735
Contingency (15%)	48,707	Mixer	₹13,740	Total	16,00,000
Prototype total	₹3,73,418	target ₹15,000 · under by ₹1,200			

Hardware ₹4.48L · Iteration ₹2.08L · People ₹4.08L

People 28% · Hardware 28% · Iteration 14% · IP + reserve 12% · Lab + tooling 9% · Software 8%

Notes: PROTOPUS cost report, 29 Sep 2020. All BOM lines, B from live listings. USD at ₹80. Imports carry 30% duty + courier. 15% contingency. Other lines, PoC Budget sheet in the costing workbook.

On screen

Where the ₹16 lakh goes, line by line

Time

30 seconds (or skip)

The one thing they must remember

Every rupee is traceable: 96 BOM lines, 81 priced from live listings.

What to say

“For anyone who wants the detail, here is every line.”

“Baker’s biggest costs are motion at 80,000 and the high-temperature jig printer at 78,000. Parts come to 3.12 lakh, then import duty and a 15% contingency.”

“Mixer is 62,000 in parts, 75,000 all in.”

“And at production, Baker is projected at 1.45 lakh and Mixer at 13,740, both under target.”

Numbers to know cold

- Baker: motion ₹80,382 · jig printer ₹77,698 · electronics ₹42,985 · Zone A ₹38,270 · Zone B ₹33,058 · frame ₹22,795
- Baker parts ₹3,12,465 + duty/courier ₹12,247 (35% on ₹34,991 of imports) + contingency ₹48,707 = ₹3,73,418
- Mixer parts ₹62,212 + ₹2,872 + ₹9,763 = ₹74,847
- USD at ₹88

Questions you may get

Q Why a ₹78,000 resin printer in a PCB machine?

It prints the high-temperature placement jigs, rated to 160 °C. In production we remove it: the jigs print on Baker’s own UV engine, which we still have to test.

Q Why a Raspberry Pi in the prototype?

Speed of development. Production uses a BTT Manta M8P board with a CB1 module instead, about ₹10,800.

Q How do you get from ₹3.73L to ₹1.45L?

Four design moves: build the axes from ball screws and rails (about ₹5,500 an axis instead of ₹16,500), one controller board instead of a Pi plus drivers, one 24 V supply, and jigs printed on Baker itself. Plus China volume pricing.

Slide 14 · 13 · What's next

13 · WHAT'S NEXT			
After proof of concept: raise, build in China, run lean from Delhi			
THE ROUND	THE CHINA PLAN	OPERATIONS	
~₹1.5 crore seed, at proof of concept	Design in Delhi, build in Shenzhen	Delhi finalises, it doesn't build	
China pilots + first batches	• Mixer: full turnkey from batch one.	• Every unit: scan, test, activate	
China set-up, IP	• Baker: China modules, final build in Delhi; turnkey once ~ \$1,305.	Proto, ship.	
Team of 5, 12 months	• 12 factories quoted + 2 pilots + 1 primary + backup.	• Mixer is our production tester: our rig, limits, data.	
Delhi centre + capital	• Pilots: 20/50/100 Mixers, 3/5/20 Bakers.	• Hire: a mechanical + a manufacturing/quality engineer.	
Buffer		• IEC, customs broker, China trademarks first.	
Seed round			
~₹1.5 Cr			
Pilot batches are sellable (~₹95L at list). Schenck interested at PoC.			
MONTH 0-2	MONTH 3-5	MONTH 6-9	MONTH 10-12
Design freeze, 12 RFQs out	Pilots in China: EVT and DVT	PVT: first 100 Mixers, 20 Bakers	Mass production, first container

On screen

After proof of concept: raise, build in China, run lean from Delhi

Time

45 seconds

The one thing they must remember

After the PoC: a ~₹1.5 crore seed, pilots in China, and Delhi as a lean finalisation centre.

What to say

"After the proof of concept, we raise, build in China and run lean from Delhi."

"The round is an indicative 1.5 crore seed: 70 lakh for China pilots and first batches, costed from our BOM, plus set-up, a small team, a Delhi centre and a buffer. The pilot batches themselves are sellable, about 95 lakh at list price."

"The China plan: we design in Delhi and build in Shenzhen. Mixer goes full turnkey from the first batch. Baker starts as tested modules from China with the final build in Delhi, and moves fully to China once the factory price makes sense."

"Operations: Delhi finalises, it doesn't build. Every unit is scanned, tested, activated and shipped, and Mixer is our production tester."

Numbers to know cold

- Seed ~₹1.5 Cr: pilots ₹70L · set-up + IP ₹25L · team of 5 ₹25L · Delhi centre ₹20L · buffer ₹10L
- Pilots: 20/50/100 Mixers, 3/5/20 Bakers · pilot batches ≈ ₹95L at list
- Baker turnkey breaks even at a factory price ≤ \$1,305 (parts alone ≈ \$1,302); Mixer ≤ \$127
- Months from the round: freeze + RFQs (0-2) · pilots (3-5) · first 100 Mixers + 20 Bakers (6-9) · first container (10-12)

Questions you may get

Q Why China and not India? What about Make in India?

Design, IP, testing, final acceptance and activation all stay in India. China is where standard parts and volume electronics assembly are cheapest today. We will quote Indian EMS partners too, and move work to India as volumes and incentives make sense.

Q How do you protect IP in China?

Non-disclosure agreements written under Chinese law, trademarks registered in China early, locked firmware only at the factory, Proto activated in India, and no single supplier builds everything.

Q Where does \$1,305 come from?

Factory price plus 16.5% duty, plus about ₹8,000 freight and ₹3,000 of India checks, must stay under today's ₹1,44,842. Parts alone are about \$1,302, so Baker turnkey only wins with volume component pricing. That is why Baker starts as modules.

Q Is ₹1.5 crore firm?

No, indicative. The pilot line is costed from the BOM; the others are estimates we will firm up during the proof of concept.

Slide 15 · Thank you



On screen

Thank you. Come and try the loop.

Time

15 seconds

The one thing they must remember

Come and try the loop: invite them to the lab.

What to say

“Thank you. I’m Soham. I cover industrial design, mechanical architecture, embedded electronics and machine software, and my first hire is a software engineer who also writes firmware.”

“Come and try the loop: I would love to show you Baker running in the lab.”

Delivery tip

End with an invitation, not a question. Then stop talking and smile.

Numbers to know cold

- protopus-kappa.vercel.app · soham9763015@ece.du.ac.in

Questions you may get

Q You are a solo founder and a student. Can you do this?

The core skills of the product are mine: mechanical, electronics and machine software. The first hire covers software and firmware. The milestones are monthly and visible, so you see the progress, not just hear about it.

Q What happens around exams and graduation?

Answer honestly with your plan and your time commitment.

Q How can we help beyond money?

Have an answer ready: introductions to maker studios and tinkering labs, manufacturing mentors, and lab space.

Tough questions bank

Questions not tied to one slide. Answer in two or three sentences, then stop.

Q Is a 10 W laser safe for students?

The laser is a 10 W, 445 nm module (Class 4 power) inside an enclosure with door interlocks, an E-stop and an orange laser-safe viewing window. OD6+ goggles are in the BOM. Safety classification of the finished machine is part of our compliance work.

Q What about chemicals and fumes?

The developer is dilute sodium carbonate, a few millilitres on a squeegee, with no tanks or baths. Milling dust goes to a vacuum. Reflow uses low-temperature paste. We budget ₹50,000 for lab safety.

Q Isn't low-temperature bismuth solder brittle?

Sn42Bi58 melts at 138 °C. It keeps the board flat and the process cool, which suits prototypes. A standard alloy can be used later where needed, since the jig lifts off before reflow.

Q What is your unfair advantage?

We close the whole loop on one desk, and Proto connects the design to real measurements, so every Mixer test makes Proto better. And the machines are designed to be built cheaply: ₹1.45 lakh projected for Baker.

Q What are the biggest risks?

Three: the four Baker process points (tested in month 3), supply of hobby-market boards (freeze revisions, keep spares), and thin cost headroom (volume quotes next). Name them before they ask.

Q How will you sell after the PoC?

Maker studios first for visibility, then direct to hobbyists and startups, and alongside, labs that already buy equipment.

Q Service and repairs?

Delhi is our finalisation and support centre, and Mixer doubles as the service tester. Most parts are standard: motors, rails, supplies.

Q Certifications?

Before selling at volume, we classify the product for laser and electrical safety and for the Wi-Fi on the CB1 module, and check BIS and e-waste rules.

Q Why are you the right person?

I built Baker's prototype and designed the whole system across mechanical, electronics and software. I live this problem in my own lab.

Q What if you don't get this funding?

I keep building at a slower pace and apply to other programmes. The milestones stay the same; only the speed changes.

The honest-answer formula

When you don't know: "I don't know yet. Here is how we'll find out, and by when." Never guess a number.

Numbers cheat sheet

Topic	Number	Status
The ask	₹16,00,000 for 9 months (6 to PoC + 3 to raise)	Plan
Baker prototype	₹3,73,418 (parts ₹3,12,465 + duty ₹12,247 + contingency ₹48,707)	Costed
Mixer prototype	₹74,847 (parts ₹62,212 + ₹2,872 + ₹9,763)	Costed
BOM basis	96 lines, 81 priced from live listings; USD at ₹88	Costed
Baker production unit	₹1,44,842 incl. ₹12,000 Delhi assembly (target ₹1,50,000)	Projected
Mixer production unit	₹13,740 incl. ₹2,000 assembly (target ₹15,000)	Projected
Selling price	Baker ₹2,99,000 · Mixer ₹34,999 · together ₹3,34,000	Projected
Margin	Baker ~33% · Mixer ~36%	Projected
Proto plan	₹9,999 / year / workspace	Planned
Competitors	Voltera ~₹3.1L · LPKF ₹8.6–24.5L · Bantam ~₹3.1L · AD3 ~₹33,000	List prices
Market	PCB ₹6.4L cr global, ₹55K cr India (~15%/yr) · T&M ₹3.1L cr / ₹8.3K cr	Sourced
People	33M+ Arduino · 2.3M+ Hackster · 10,000 ATLS (+50,000) · ~2,500 Fab Labs	Sourced
Seed round	~₹1.5 Cr: pilots ₹70L · set-up ₹25L · team ₹25L · Delhi ₹20L · buffer ₹10L	Indicative
China break-even	Baker factory price ≤ \$1,305 (parts ≈ \$1,302); Mixer ≤ \$127	Model
₹10L fallback	Keep hardware ₹4.48L + Rev B ₹1L + ~6 months of the engineer; defer Mixer boards, IP, outreach	Contingency plan

Fix or decide before you present

Each item is something a sharp juror could notice. Settle it, then practise the answer.

#	Item	Why it matters	What to do
1	Board size: the workflow says 100 × 160 mm; the BOM buys 100 × 150 mm FR4	A technical juror may ask the board size	Pick one and use it everywhere
2	Slide 08 says “free Baker and Mixer units” now; the ₹16L budget has 5 Mixers for studios but no extra Baker	Looks like a contradiction with the ask	Use the prepared answer, or change the slide to “Mixer units now, Baker units after the seed round”
3	Baker “raw prototype running” vs ₹3.73L to build Baker	“Why pay for what already runs?”	Know exactly what the raw prototype contains and what the ₹3.73L adds
4	Proto claims (Flux.ai API, Gemini, Gerbers, waveform diagnosis)	Jurors may ask for a demo	Know which run today and which are in development; say it precisely
5	Milestone targets (20 boards, monthly demos)	They will hold you to them	Confirm you are comfortable, or adjust the slide
6	Grant rules	Some schemes cap awards or disallow some costs	Check the scheme’s cap and rules; have the ₹10L fallback ready
7	Company status and IP	A common eligibility question	Know whether you are incorporated and what IP is filed

Last check, the night before

- Run the full pitch twice with a timer; cut words, not slides.
- Answer the tough-questions bank out loud once.
- Charge the laptop, download the demo video, keep the deck as a PDF on your phone.
- Sleep. Energy and clarity beat one more rehearsal.